



Research article

An integrated simulation method for process window determination and deformation analysis in gas–liquid low-pressure forming of complex thin-walled tubes

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ABSTRACT

Gas–liquid low-pressure tube forming (GLLF) suppresses compressive instability and improves corner filling in complex thin-walled tubular components. However, effective methods are still lacking to analyze volume–pressure coupling and clarify how pressure response affects instability suppression, deformation behavior, and axial/circumferential forming mechanisms. To address these issues, a gas–liquid compression model was embedded into finite element analysis, and a volume–pressure coupled method was developed by integrating ABAQUS restart analysis with Python scripting. Based on this method, the process window was determined, and experiments were conducted to validate the influence of pressure evolution on instability suppression and deformation behavior. At an initial pressure of 3 MPa, reducing the gas-phase volumetric fraction from 0.80 to 0.28 increased the terminal pressure from approximately 4 MPa to 18 MPa and reduced the maximum corner deviation from about 2.6 mm to below 0.05 mm. Stress, cross-sectional shape, and wall-thickness analyses revealed the bending–compression dominated corner-radius evolution and thickness redistribution governed by curvature variation and deformation sequence. Overall, this study provides a validated method for process design and instability control, and reveals the axial and circumferential deformation mechanisms in complex thin-walled tubes GLLF, including shape–dimension evolution and wall-thickness distribution characteristics.

1. Introduction

With the intensification of global climate change and the continuous increase in carbon emissions from the transportation sector [1], high-end transportation equipment—including aviation, aerospace, marine, and automotive systems—faces an urgent demand for lightweight and highly integrated structural solutions [2]. Increasing the proportion of integrated thin-walled structures [3] is regarded as an effective strategy to reduce energy consumption while enhancing load-carrying capacity and structural efficiency [4].

The forming methods for integrated thin-walled structures mainly include deep drawing [5], additive manufacturing [6], spinning [7], die casting [8], and hydroforming technologies [9]. Among these, hydroforming has become one of the most important techniques for manufacturing complex thin-walled tubular components. Compared with conventional stamping and welding processes, hydroforming can shorten

the processing route, reduce energy consumption, and significantly improve structural integrity and mechanical performance [10].

Existing hydroforming technologies for complex thin-walled metallic tubes can generally be categorized into two types: high-pressure hydroforming bulging [11], [12] and low-pressure forming [13], [14]. High-pressure hydroforming relies primarily on elevated internal fluid pressure to expand the tube into the die cavity, whereas low-pressure forming utilizes the coordinated interaction between die motion and internal pressure to complete deformation [15].

In high-pressure hydroforming, the forming force is mainly provided by high-pressure internal fluid. Although this process improves die filling, complex thin-walled components are prone to localized thinning or even cracking during tensile deformation, especially in transition fillet regions, which weakens the structural strength of the final part [16]. Therefore, key challenges include optimizing material distribution during preforming [17] and controlling the internal pressure

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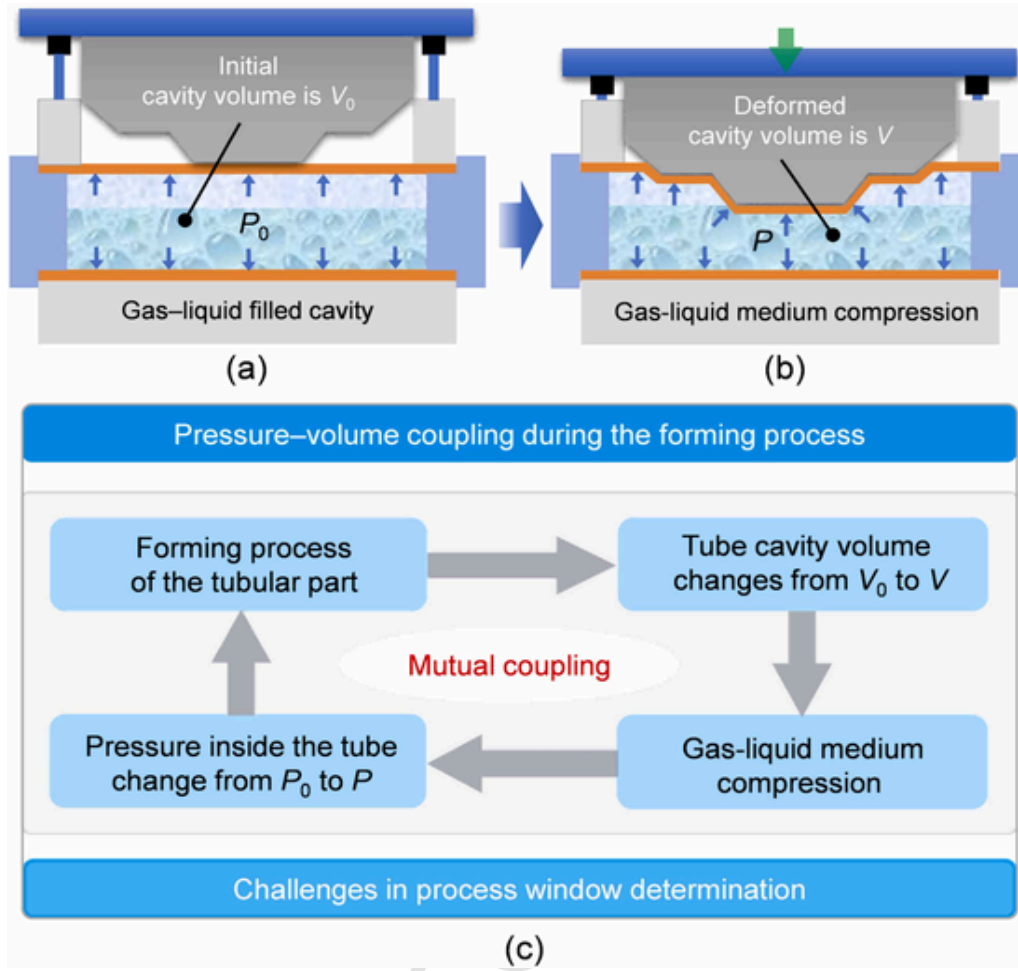


Fig. 1. Volume–pressure coupling in tube GLLF: (a) Initial cavity volume V_0 and internal pressure P_0 ; (b) Deformed cavity volume V and updated internal pressure P caused by gas–liquid mixture compression; (c) Closed-loop coupling mechanism among tube deformation, cavity volume variation from V_0 to V , gas–liquid mixture compression, and pressure evolution from P_0 to P .

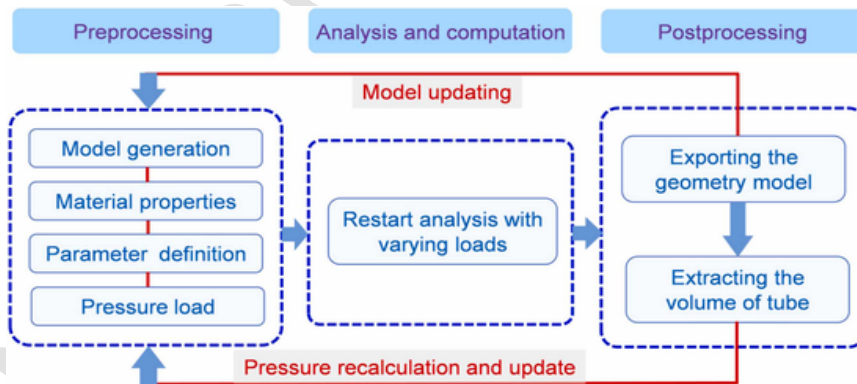


Fig. 2. Finite element simulation procedure for tube GLLF with restart-based model updating and pressure-load updating for volume–pressure coupling analysis.

loading path [18]. To address these issues, intelligent loading-path control strategies have been developed. Manabe et al. [19] monitored and adjusted the pressure–displacement curve in real time to improve forming efficiency and stability. Li et al. [20] proposed a fuzzy control strategy to optimize loading paths and suppress wrinkling and cracking. Studies on wrinkling behavior and controlled wrinkling have further

provided guidance for instability control and formability improvement in tube hydroforming [21], [22].

As tubular components become increasingly complex, involving variable diameters, non-uniform cross-sections [23], spatially curved axes [24], or small corner radii [25], multiple forming steps such as bending, diameter reduction and pressing are often required, thereby

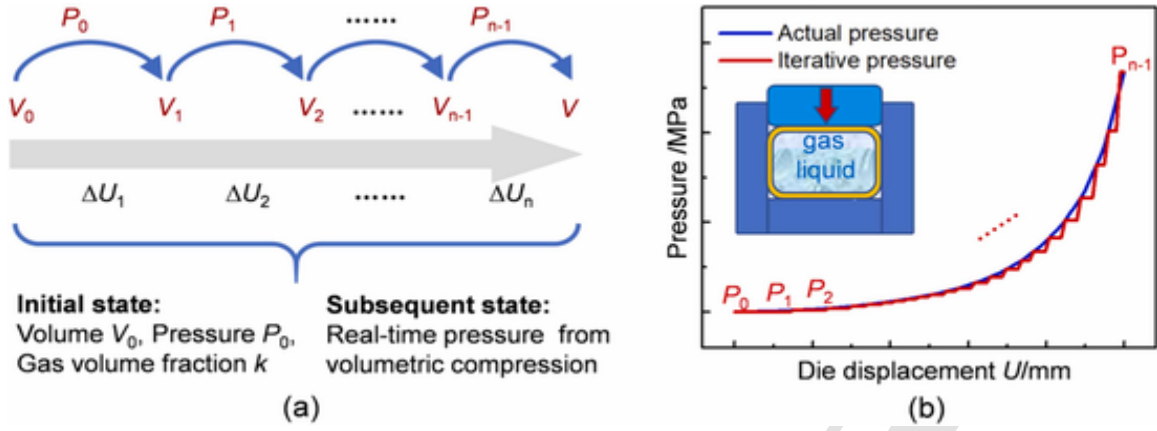


Fig. 3. Schematic of volume–pressure evolution and iterative pressure determination: (a) Evolution of cavity volume and internal pressure during forming as die displacement U changes (V_0 – V_n , P_0 – P_{n-1}); (b) Comparison between actual pressure and iterative pressure as a function of die displacement, with schematic illustration of the tube GLLF.

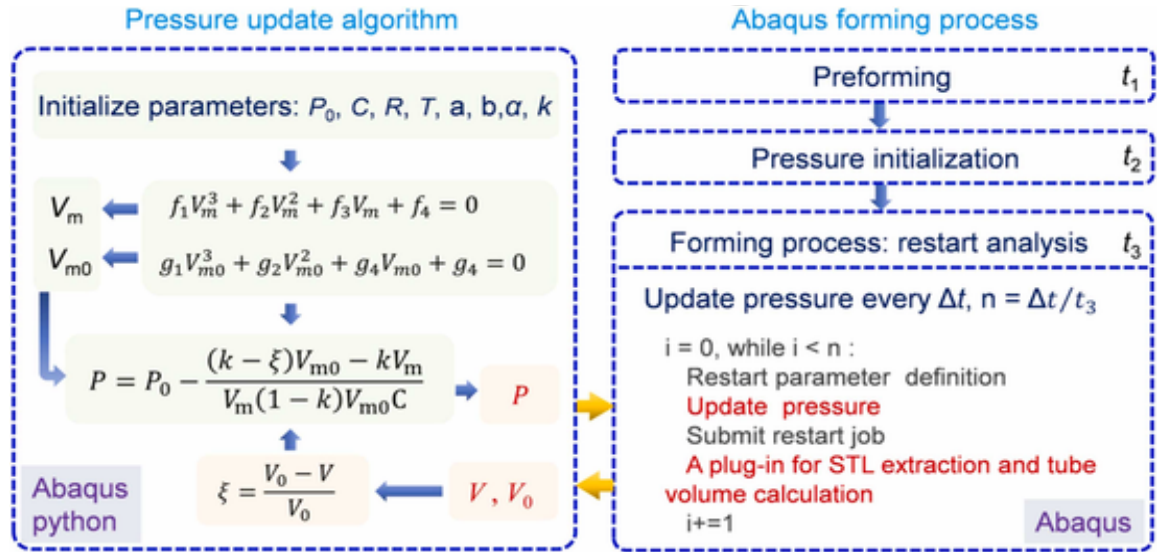


Fig. 4. Pressure updating algorithm for the SRK-based gas–liquid mixture compression model via ABAQUS restart analysis. At each restart increment Δt , the tube cavity volume is extracted from STL files using a custom-developed plugin, the compression ratio ξ is calculated, and the internal pressure P is updated through ABAQUS Python scripting, enabling automated volume–pressure coupling.

increasing process complexity. Accordingly, low-pressure forming of thin-walled tubes has emerged as a cost- and energy-efficient alternative to high-pressure hydroforming [26]. In this process, die motion provides the primary forming force, while the internal medium offers auxiliary support, converting tensile-dominated deformation into compression-dominated deformation [27]. Compared with hydroforming, low-pressure forming reduces forming pressure, equipment tonnage and energy consumption [28], while enabling the formation of smaller local fillets at room temperature under relatively low pressures [29]. These advantages indicate its potential for efficient and sustainable manufacturing. Currently, the main low-pressure forming methods include liquid-filled pressing [30], liquid-filled upsetting [31] and liquid impact forming [32].

Compressive deformation introduces additional challenges. Under compressive stress states, materials are prone to instability and wrinkling [33], [34], similar to circumferential compressive instability in deep drawing [35]. Suppressing such defects requires sufficient normal support pressure [36]. In low-pressure tube forming, sidewall compres-

sion shows similar behavior. Chu et al. [37] proposed a plasticity-based buckling pressure model for compressive instability in thin-walled tubes. Pressure loading strategies, including linear ramps, constant pressure, and pulsating paths, have been investigated to improve thickness uniformity and formability [38], [39]. Zhang et al. [40] developed a multidirectional hydraulic calibration method combining internal pressure and axial compression to improve geometric precision in additively manufactured tubes. Sun et al. [29] analyzed thickness variation in axial hydro-forming of variable-diameter tubes and clarified defect mechanisms.

Despite these advances, studies on internal-pressure-supported axial compression have not systematically addressed forming processes involving axial geometric features. Moreover, in conventional low pressure forming, the compressibility of the liquid medium leads to volume-induced pressure fluctuations. The limited flexibility of traditional loading strategies and the difficulty in maintaining ideal pressure states result in reduced forming efficiency, restricting large-scale industrial application [41].

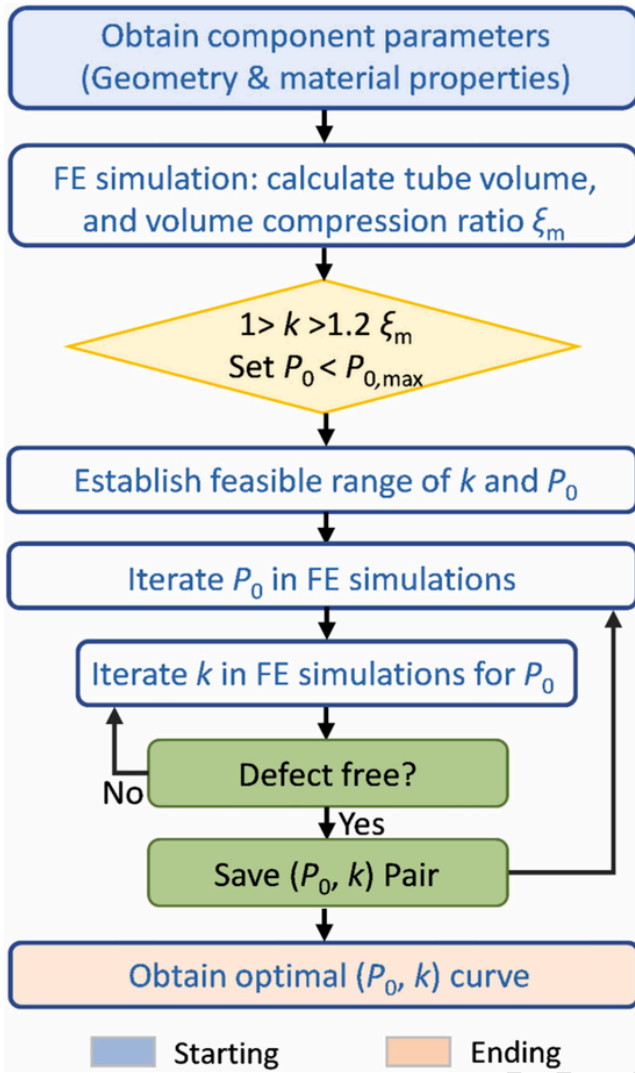


Fig. 5. Flowchart of process window determination for tube GLLF, including component parameter acquisition, volume compression ratio calculation, feasible range establishment for P_0 and k , iterative FE simulation, and final determination of the (P_0, k) curve.

Gas-liquid low-pressure forming (GLLF) has been proposed to overcome these limitations [42]. By combining the high compressibility of gas with the near incompressibility of liquid, a hybrid medium with tunable compressibility can be created to provide controllable internal support. Favorable pressure-evolution curves can be attained by controlling the gas-liquid ratio and initial pressure, thereby effectively suppressing instability and wrinkling while enabling the formation of smaller local corner features under relatively low support pressures [43].

Although GLLF improves formability under low pressures, several gaps remain. The volume-pressure coupling caused by cavity volume change and gas-liquid mixture compression has not been adequately captured in finite element simulations. Moreover, the mechanisms of circumferential compression instability, axial corner filling, and material flow remain unclear, especially for complex thin-walled tubes with local geometric features. As a result, reliable forming-window determination still lacks a systematic design method and largely depends on empirical trials.

This study develops a finite element-based volume-pressure coupled method for GLLF to address these gaps and clarify the coupling

among volume evolution, gas-liquid mixture compression, pressure response and material flow. A gas-liquid mixture compression model is integrated into ABAQUS through Python scripting and restart analysis, enabling pressure updating according to instantaneous cavity volume evolution. An iterative strategy is then proposed for process parameter determination, instability suppression and axial corner filling improvement. The deformation behavior is further analyzed by examining stress evolution, shape development and the factors affecting circumferential and axial wall thickness distributions. Experiments on complex AA6061 thin-walled tubes are conducted to validate the method and analysis.

2. An integrated simulation method for tube GLLF

2.1. Volume-pressure coupling during tube GLLF

In tube GLLF, the evolution of tube cavity volume and internal pressure exhibits an instantaneous and strongly coupled interaction. As illustrated in Fig. 1, plastic deformation of the tube changes the cavity volume from the initial value V_0 to the deformed value V . The variation in cavity volume compresses the gas-liquid mixture, thereby changing its compression state and driving the real-time evolution of internal pressure from P_0 to P . Conversely, the updated pressure directly regulates the deformation behavior of the tube, further affecting the cavity volume during subsequent forming. Therefore, the forming process, cavity volume variation, gas-liquid mixture compression, and pressure evolution are interdependent and constitute a closed-loop dynamic feedback mechanism that governs the entire forming process.

However, conventional finite element approaches typically treat internal pressure as a predefined loading condition, making it difficult to achieve real-time bidirectional updating between volume and pressure. As a result, the coupled evolution between medium compression and structural deformation cannot be accurately captured. Therefore, developing a simulation method capable of dynamic volume-pressure coupling is essential for achieving reliable numerical prediction of tube GLLF.

2.2. Dynamic solution of volume-pressure coupling

To achieve dynamic volume-pressure coupling in tube GLLF, a time-discretized, volume-driven pressure updating strategy is proposed. The forming process is divided into incremental stages, where deformation is solved under constant internal pressure. After each stage, the current tube volume is extracted to calculate the compression ratio, and the pressure is updated using the gas-liquid compression model. The updated pressure is then reassigned for restart analysis, forming an iterative volume-pressure feedback loop.

As illustrated in Fig. 2, the finite element simulation method consists of three modules: preprocessing, analysis and computation, and postprocessing. The core concept lies in combining stage-wise constant-pressure computation with volume-based pressure updating under a time-discretized scheme, thereby enabling dynamic coupling between structural volume evolution and internal pressure response.

The proposed approach discretizes the continuous forming process into n incremental time steps, within which the internal pressure is recalculated and updated according to the structural volume evolution, thereby enabling synchronized evolution of pressure and deformation. As illustrated in Fig. 3, the initial pressure is P_0 , corresponding to the initial volume V_0 . When deformation causes the tube volume to update to V_i , the updated volume V_i is entered the gas-liquid compression model to compute a new pressure P_i , which is then applied as the boundary condition for the current increment. Repeating the iterative procedure of volume update-pressure computation-pressure application generates a discrete pressure evolution path as a function of volume change. As the time increment Δt decreases, the discrete numerical

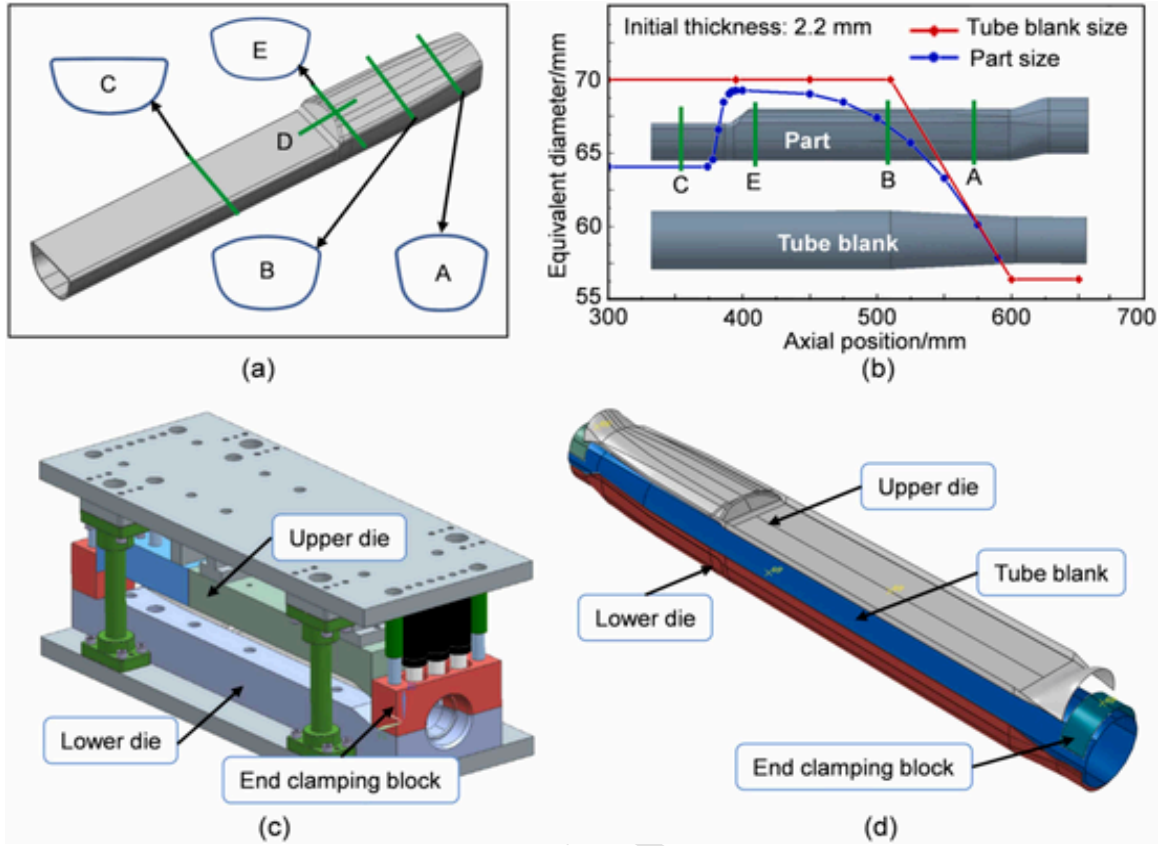


Fig. 6. Geometry dimension analysis and forming model of the complex thin-walled tube: (a) Geometric model of the complex tube with typical cross-sections A-E; (b) Comparison of axial contour profiles between the initial tube blank and the formed part; (c) Geometric model of the forming die assembly, including the upper die, lower die, and end clamping blocks; (d) ABAQUS finite element model of the tube GLLF.

pressure curve converges toward the theoretical continuous pressure response.

2.3. Modeling and implementation of volume–pressure coupling

2.3.1. Theoretical model of gas–liquid mixture compression

Our previous study [42] systematically elucidates the fundamental principles of tube GLLF and establishes a theoretical calculation model for gas–liquid mixture compression based on the Soave–Redlich–Kwong (SRK) equation of state. In developing the model, gas dissolution in the liquid phase is considered [44], enabling a more accurate description of the nonlinear compressive behavior of the gas–liquid mixture under pressure. An isothermal assumption is adopted in the derivation because the forming process is conducted at room temperature under low-pressure conditions, and both the tube and die have relatively large heat capacities. Consequently, temperature variation of the gas–liquid mixture during compression is limited and can be neglected. The results demonstrated that the proposed model can reliably predict the pressure evolution of the medium during the forming process.

According to the SRK real gas equation of state [45], the gas–liquid compression model assumes an initial pressure P_0 and a gas-phase volumetric fraction k . When the gas–liquid mixture undergoes compression, and the volumetric compression ratio is ξ , the medium volume changes relative to its initial state. Under this condition, the internal pressure P can be expressed as:

$$P = P_0 - \frac{(k - \xi) V_{m0} - k V_m}{V_m (1 - k) V_{m0} C} \quad (1)$$

In Eq. (1), V_m represents the gas molar volume at the current state, while V_{m0} denotes the molar volume at the initial state. The parameter C corresponds to the dissolved gas quantity in a unit volume of liquid. According to Eq. (1), determination of the current pressure P requires simultaneous evaluation of the current molar volume V_m and the initial molar volume V_{m0} .

The initial molar volume V_{m0} is evaluated based on the SRK gas equation of state. Introducing the initial temperature, initial pressure, and gas-specific thermophysical parameters into the SRK equation gives a cubic expression for the gas molar volume:

$$P_0 V_{m0}^3 - RT V_{m0}^2 - (P_0 b^2 + RTb - \alpha a) V_{m0} - \alpha ab = 0 \quad (2)$$

Where R is the gas constant, T denotes the temperature, a and b are gas-specific parameters related to the critical properties, and α is the correction factor. For numerical convenience, Eq. (2) can be rearranged into the standard cubic polynomial form:

$$g_1 V_{m0}^3 + g_2 V_{m0}^2 + g_3 V_{m0} + g_4 = 0 \quad (3)$$

Where:

$$\begin{cases} g_1 = P \\ g_2 = RT \\ g_3 = -(P_0 b^2 + RTb - \alpha a) \\ g_4 = \alpha ab \end{cases} \quad (4)$$

The volumetric compression ratio ξ for the tube is expressed as:

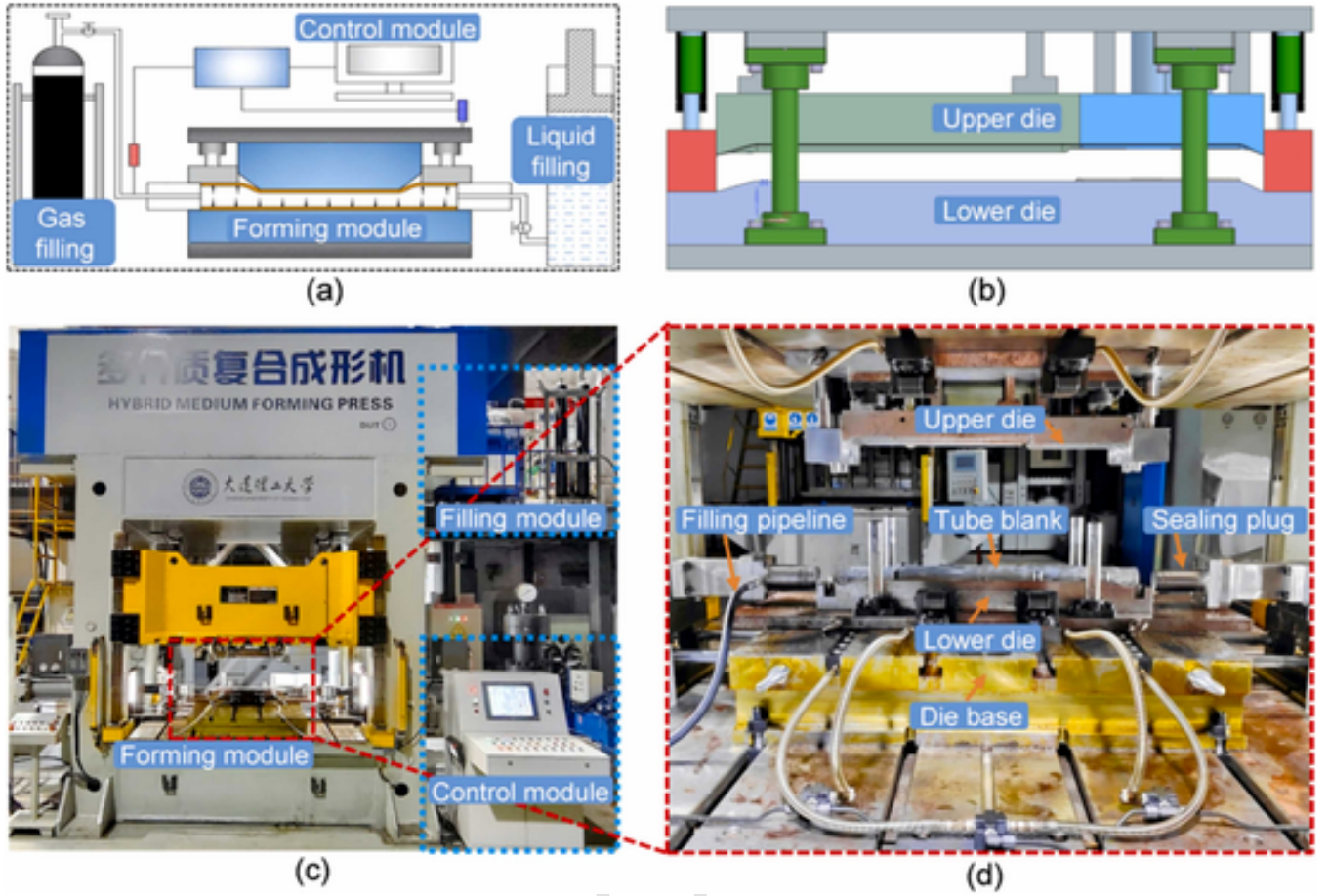


Fig. 7. Experimental setup and die structure for GLLF of the complex thin-walled tube: (a) Schematic diagram of the GLLF system, including gas/ liquid filling, control module, and forming module; (b) Structural configuration of the upper and lower dies; (c) Photograph of the hybrid medium forming press and overall equipment layout; (d) Detailed view of the forming area, including upper die, lower die, tube blank, sealing plug, filling pipeline, and die base.

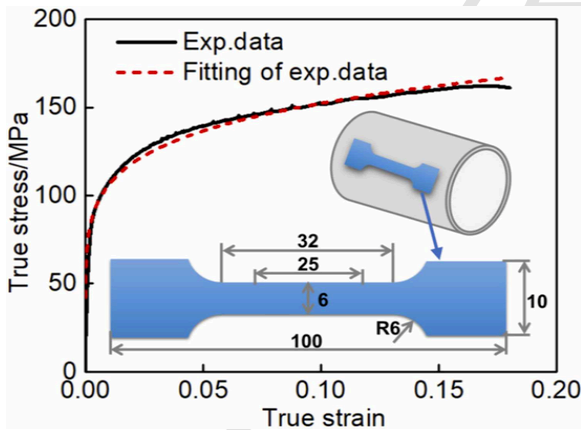


Fig. 8. True stress-strain curve of AA6061.

Table 1
Material properties of AA6061 tube.

Yield strength σ_0 (MPa)	Young's modulus E (GPa)	K (MPa)	n
82.8	69	218.3	0.156

$$\xi = \frac{V_{m0} - V_m}{V_{m0}} \quad (5)$$

Eq. (5) establishes the relationship between the current gas molar volume V_m and the initial molar volume V_{m0} . Accordingly, V_{m0} can be expressed in terms of V_m and the compression ratio ξ , and substituted into Eq. (1) to eliminate the initial molar volume term. As a result, the pressure expression becomes solely dependent on the current molar volume. By further coupling this relation with the SRK equation of state Eq. (2), the pressure expression can be rewritten as a cubic polynomial in V_m :

$$\left[P_0 + \frac{k}{C(1-k)V_{m0}} \right] V_m^3 - \left[\frac{k-\xi}{C(1-k)} + RT \right] V_m^2 + \left[a\alpha - bRT - P_0b^2 - \frac{kb^2}{C(1-k)V_{m0}} \right] V_m + \frac{k-\xi}{C(1-k)}b^2 - ab\alpha = 0 \quad (6)$$

For numerical implementation, Eq. (6) is reformulated into the standard cubic polynomial form, where the coefficients are defined as follows:

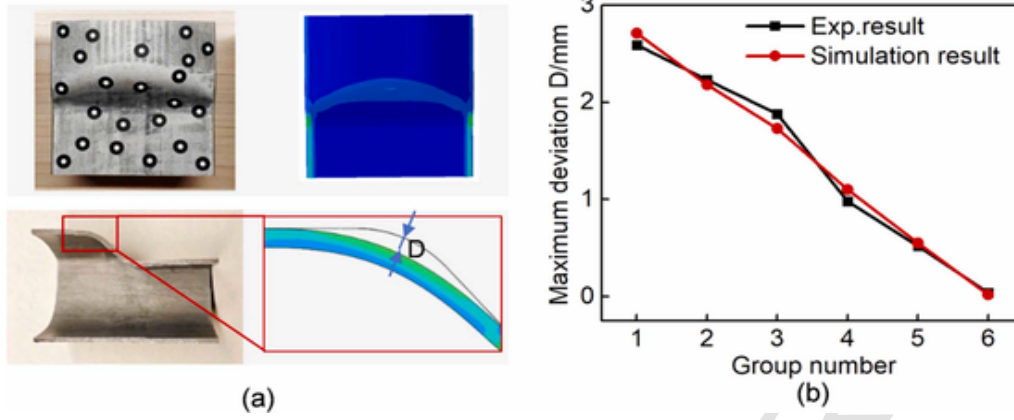


Fig. 9. Validation of the maximum fillet deviation D at $P_0 = 3$ MPa with different $k = 0.80\text{--}0.28$: (a) Experimental and simulated results with the definition of D ; (b) Comparison between experimental and simulated deviations.

$$\begin{cases} f_1 = P_0 + \frac{k}{C(1-k)V_{m0}} \\ f_2 = -\frac{k-\xi}{C(1-k)} - RT \\ f_3 = a\alpha - bRT - P_0b^2 - \frac{kb^2}{C(1-k)V_{m0}} \\ f_4 = \frac{k-\xi}{C(1-k)}b^2 - a\alpha \end{cases} \quad (7)$$

Therefore, Eq. (6) can be expressed in the standard cubic polynomial form as:

$$f_1V_m^3 + f_2V_m^2 + f_3V_m + f_4 = 0 \quad (8)$$

Eq. (8) establishes a nonlinear cubic relation between the current gas molar volume V_m and the volumetric compression ratio ξ . Solving the equation yields the physically meaningful molar volume V_m , while the initial molar volume V_{m0} is derived from Eq. (3). Substituting V_m and V_{m0} into Eq. (1) provides the updated internal pressure P corresponding to the given compression state.

2.3.2. Finite element implementation of tube GLLF

A volume-driven pressure updating algorithm was developed based on the ABAQUS restart analysis technique to achieve dynamic coupling between the gas-liquid mixture compression model and the finite element solution process. As illustrated in Fig. 4, the continuous forming process is discretized into multiple restart stages. In the initial stage, the initial gas molar volume V_{m0} is determined from the initial pressure using the SRK equation of state. At the end of each restart stage, the deformed tube geometry is exported from the ODB file as an STL model, and the current cavity volume is measured using a custom-developed plugin. The corresponding compression ratio ξ is then calculated. The gas-liquid mixture compression model is coupled with the equation of state to solve the current gas molar volume V_m . Based on V_m , V_{m0} , and ξ , the updated internal pressure is obtained and applied as the boundary condition for the subsequent restart step.

The procedure is executed at a time interval Δt , forming a closed-loop iterative scheme consisting of “volume extraction-compression ratio evaluation-molar volume solution-pressure update-model restart.” A smaller Δt , or equivalently a larger number of pressure updates ($n = t_3/\Delta t$), improves the approximation of the discrete pressure path to the continuous theoretical response. In this study, Δt was determined through convergence analysis by gradually increasing n until the variation in the pressure response between two successive refinements be-

came negligible. This ensures sufficient numerical accuracy while avoiding excessive computational cost.

ABAQUS secondary development is employed to extract the instantaneous tube volume and redefine the internal pressure at each increment. Combined with restart analysis, a stepwise iterative computation is realized, establishing a dynamically coupled simulation method between gas-liquid mixture compression behavior and structural deformation. Through this iterative updating mechanism, dynamic finite element coupling between volume variation and pressure response of the gas-liquid mixture is achieved, providing a stable and reliable computational method for accurate simulation of tube GLLF.

3. Determination of the process window in GLLF

3.1. Adjustment strategy for process parameters

In tube GLLF, internal pressure evolution is primarily governed by the initial gas-phase volumetric fraction k , the initial pressure P_0 , and the instantaneous volumetric compression ratio ξ of the tube. Under identical conditions, a higher P_0 or a lower k results in greater internal supporting pressure. For a given tubular component, the maximum volumetric compression ratio ξ_m is mainly determined by its geometric characteristics and deformation mode, and can therefore be regarded as approximately fixed. Consequently, the core objective of process adjustment is to determine the feasible range and matching relationship between k and P_0 .

The parameters determination procedure is illustrated in Fig. 5. First, the fundamental component parameters, including geometric dimensions and material properties, are obtained. Finite element simulations are performed to evaluate the initial tube volume V_0 and the final formed volume V , from which the maximum volumetric compression ratio ξ_m is determined. To ensure enough compressible space for the mixture during forming, the k is set to satisfy:

$$1 > k > 1.2\xi_m \quad (9)$$

The coefficient 1.2 is introduced to provide a 20% safety margin above the calculated maximum volumetric compression ratio, thereby ensuring sufficient compressible space for the gas-liquid mixture. This prevents excessive gas compression, which would otherwise cause a sharp pressure rise and compromise process stability.

Meanwhile, for a tube blank with yield stress σ_0 , wall thickness t , and initial radius R_0 , the initial pressure is limited by an upper bound to prevent bulging, given by:

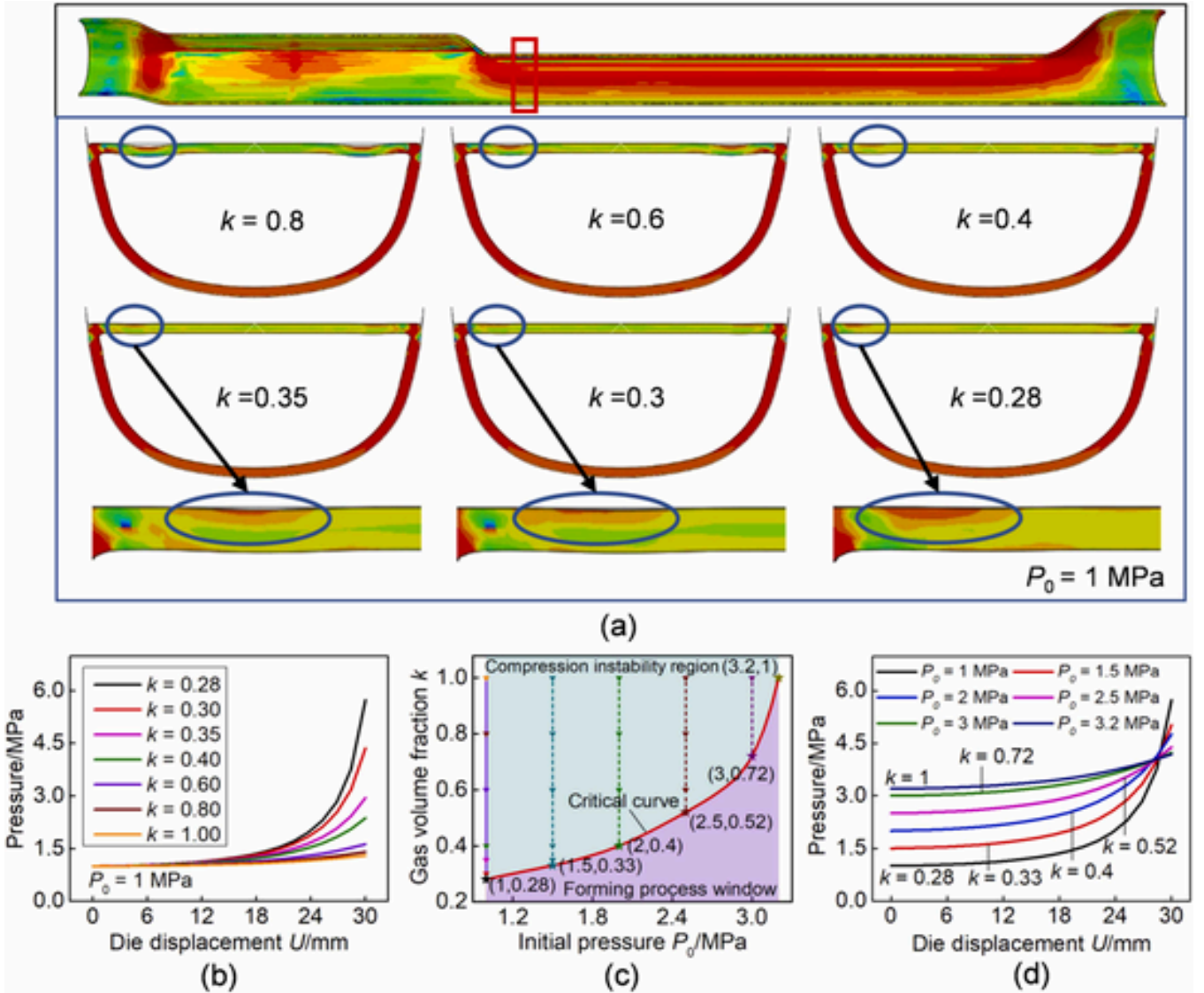


Fig. 10. Process parameter determination for suppression of circumferential compressive instability: (a) Cross-sectional deformation under different gas-phase volumetric fractions k at an initial pressure P_0 of 1 MPa, highlighting wrinkling behavior; (b) Pressure evolution curves corresponding to different k ; (c) Critical relationship between P_0 and k for wrinkling suppression, indicating the compressive instability zone and forming process window; (d) Pressure curves under different critical P_0 and k .

$$P_{0,\max} = \frac{\sigma_0 t}{R_0} \quad (10)$$

The feasible ranges of k and P_0 are first established. Uniform sampling is then performed within the range of P_0 , with $P_{0,\max}$ set as the initial value and the iteration interval and lower bound defined accordingly. For each P_0 , k is iteratively evaluated using FE simulations of tube GLLF. The admissible k is determined based on quantitative defect criteria, including circumferential height deviation (instability assessment) and maximum fillet deviation D (corner filling quality). The process window boundary is obtained by satisfying both constraints, i.e., no wrinkling and acceptable fillet deviation. This establishes a process window determination method based on coupled theoretical modeling and FE simulation.

3.2. Finite element model of the complex thin-walled tube

A complex tubular component from an electric-assisted bicycle frame was selected to investigate tube GLLF. The component is a typical thin-walled tube with variable diameter and non-uniform cross-sections, and its overall geometry and representative cross-sectional features are illustrated in Fig. 6. Based on these geometric characteristics, the cross-section types and key dimensional parameters were systematically analyzed to determine a suitable forming route and the initial tube blank size.

Tube GLLF primarily relies on circumferential compressive deformation of the cross-section. The maximum equivalent diameter of the part is 68.6 mm. Therefore, the initial tube diameter for GLLF should be slightly larger than the final component dimension. Considering deformation requirements and forming allowance, the maximum initial tube diameter was set to 70 mm. Fig. 6b presents the variation of cross-sectional dimensions along the axial direction in the key regions of the component. Some representative cross-sections were selected for analy-

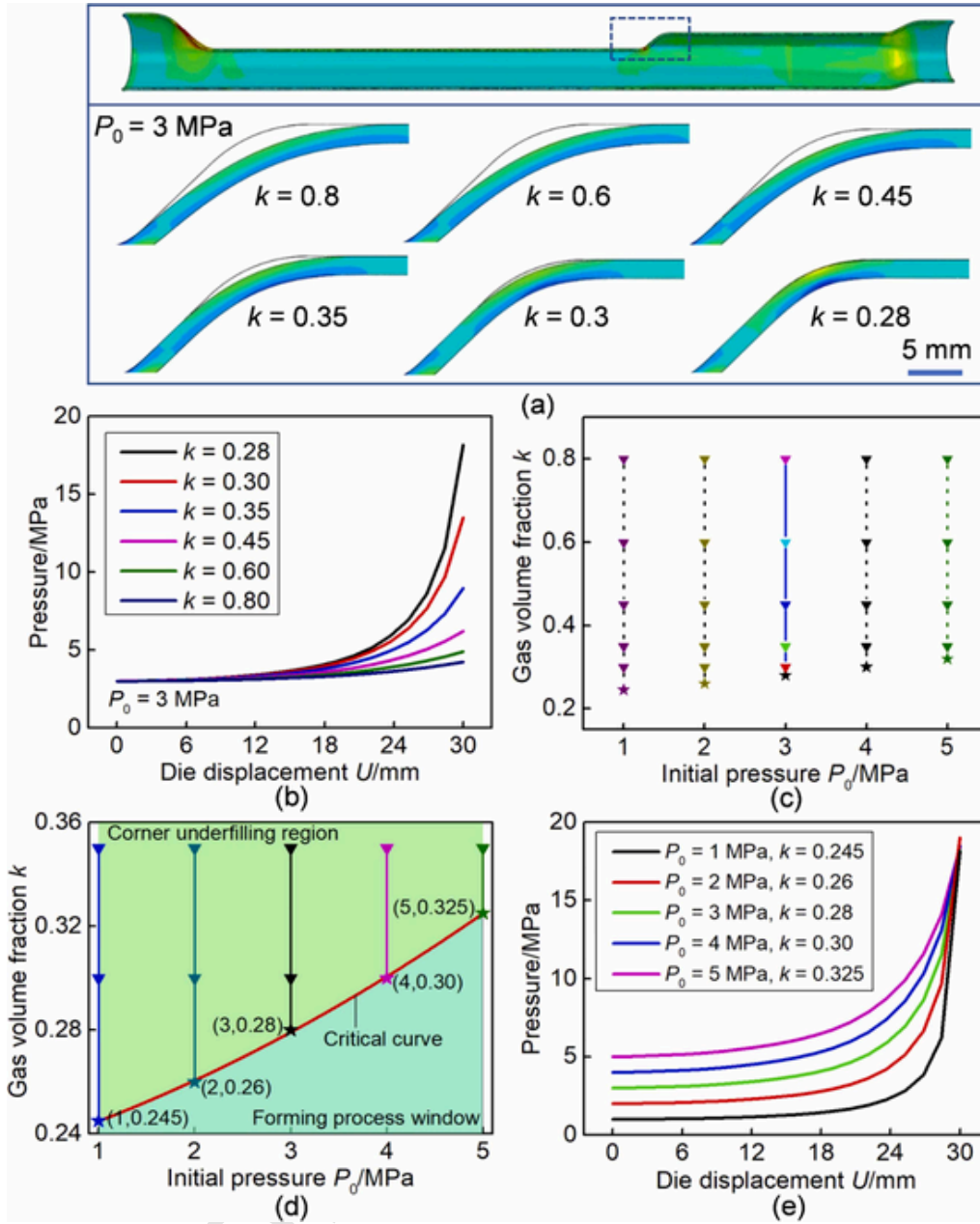


Fig. 11. Process parameters determination of GLLF for axial corner underfilling: (a) Deformation evolution under different k at $P_0 = 3$ MPa of the axial transition region; (b) Corresponding pressure evolution curves; (c-d) Critical relationship between P_0 and k for complete die fitting, indicating the corner underfilling zone and process window; (e) Pressure evolution curves under critical parameter combinations.

sis, and their dimensions and corresponding compression ratios are different. The variable-diameter region at the right end was first performed using a tube-reducing process, followed by final shaping through GLLF.

As shown in Fig. 6c, the forming system consists of upper and lower dies, end-clamping units, and sealing components, enabling the establishment of a controlled gas-liquid supporting environment. During forming, the tube blank is positioned within the die cavity and sealed at both ends. A predetermined amount of liquid and gas is then introduced into the tube, and the gas-phase volumetric fraction k is defined by controlling the gas-liquid ratio, thereby establishing an initial pressure P_0 . The dies subsequently close to induce circumferentially dominated compressive deformation, completing the shaping of the target component. As the internal cavity volume decreases, the gas-liquid mixture is

compressed, resulting in a pressure increase that provides necessary radial support, effectively suppressing compressive instability and promoting the formation of local geometric features.

Fig. 6d shows the corresponding ABAQUS FE model. The model is employed to simulate the coupled response between structural compressive deformation and pressure self-amplification, serving as a numerical validation platform for the proposed pressure updating algorithm and process parameter adjustment strategy. Explicit dynamic analysis is adopted, with a tube mesh element size of 1 mm and a die mesh element size of 1.5 mm. Coulomb friction is applied, with a coefficient of 0.1 based on the experimental lubrication conditions. Once the model parameters are defined, the full forming process is simulated via restart analysis using custom Python scripts, enabling finite element simulation of tube GLLF.

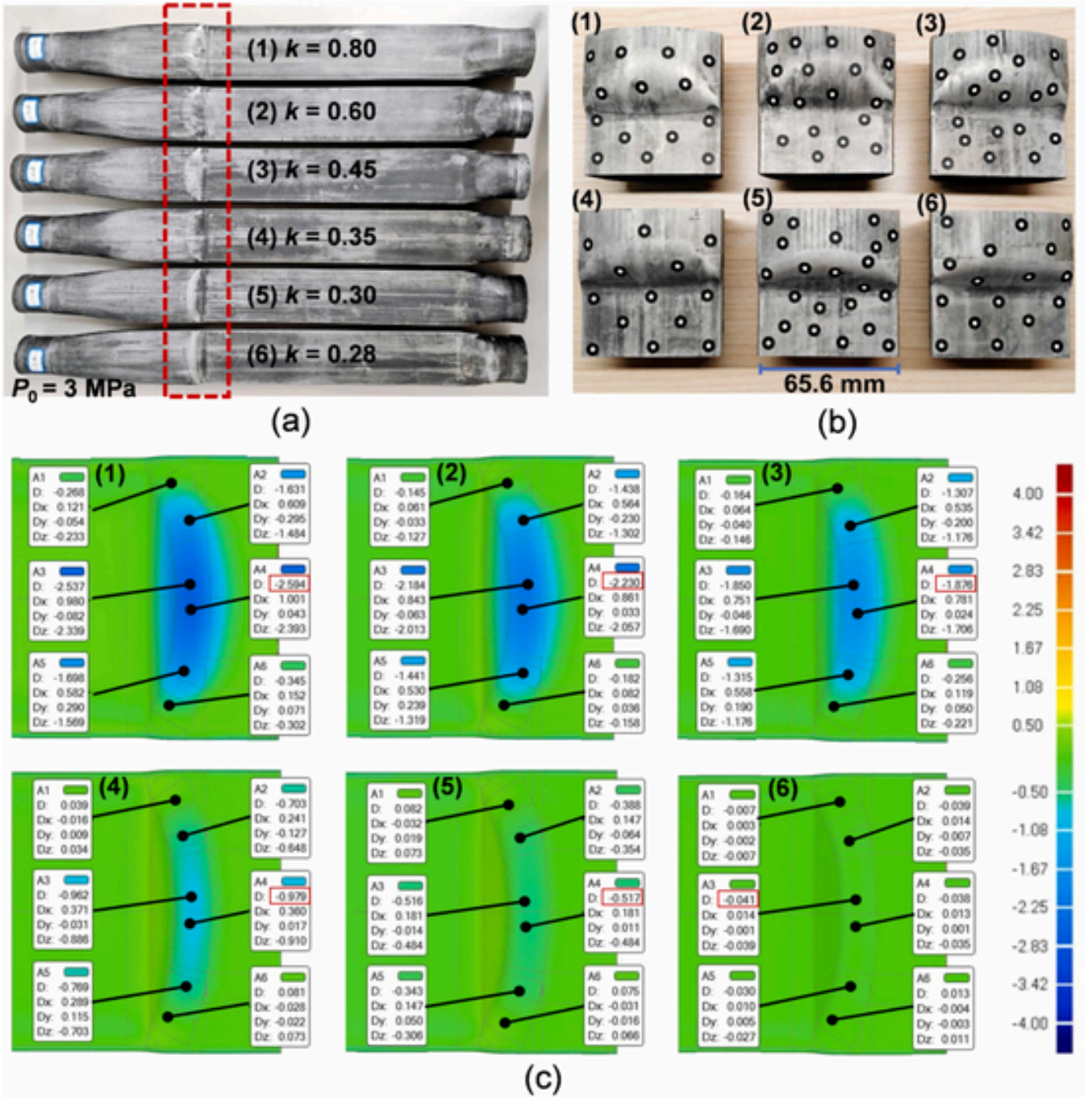


Fig. 12. Experimental results and deviation analysis of axial feature forming: (a) Formed components under different process parameters ($k = 0.80\text{--}0.28$, $P_0 = 3 \text{ MPa}$) and the selected axial region for measurement; (b) Sectioned samples of the axial transition region under different conditions (1–6); (c) Deviation distribution maps of the scanned sections corresponding to cases (1–6), showing the fitting accuracy between the formed parts and the target geometry.

3.3. Experimental equipment and specimen preparation

To validate the proposed volume–pressure coupling approach and conduct experimental research on tube GLLF, a specially designed experimental system was established. As shown in Fig. 7a, b, the schematic diagram and die model illustrate the working principle of the system. The system mainly consists of a gas/liquid supply module, a control module, and a forming module. During the forming process, the tube cavity is first filled with a gas–liquid mixture, and the tubular component is then formed by the downward movement of the die.

As shown in Fig. 7c, d, the GLLF apparatus and die assembly were developed for the experimental investigation. The gas/liquid supply module consists of a liquid-filling unit and a gas-charging unit, which are used to control the liquid-filling volume and gas pressure, respectively. The forming module includes a 200-ton servo press, a customized forming die, and horizontal sealing cylinders. The control module coordinates the sequential operation of each module to complete the forming process.

The forming procedure mainly includes three stages: (i) tube placement, die closing, and end sealing. Before the die is fully closed, the tube ends are first sealed using the end-sealing block in combination

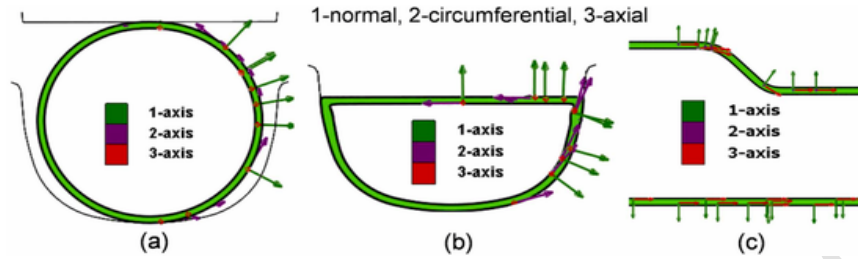


Fig. 13. Material orientation of the tube blank: (a) Initial cross-section; (b) Deformed cross-section; (c) Local view of the transition region.

with a sealing plug inserted into the inner tube wall; (ii) medium filling. A gas-liquid mixture with a predefined initial pressure is introduced into the tube cavity to establish a stable internal pressure support environment; and (iii) die closing, forming, and part removal. Under internal pressure support, the die continues to close, thereby completing the compressive forming of the tubular component.

The material used in this experiment is tube billet made of AA6061, with a maximum outer diameter of 70 mm, a length of 600 mm, and a wall thickness of 2.2 mm. Uniaxial tensile specimens were designed in accordance with ASTM E8/E8M-22 and extracted from the original tube blank. The true stress-strain curve of the material was obtained through uniaxial tensile testing, as shown in Fig. 8. The experimental data were fitted using the power-law (Hollomon) hardening model, $\sigma = K\epsilon^n$, and the key mechanical parameters are summarized in Table 1. The tensile-derived parameters were used to define the isotropic elastoplastic response in FE simulations. Although the GLLF process is dominated by circumferential compression, such data are commonly used when compressive test data are unavailable.

3.4. Validation of the finite element model

The simulation accuracy was validated by comparing the simulated and experimental axial fillet forming results. The maximum fillet deviation D , which directly reflects axial corner filling accuracy, was selected as the validation index. As shown in Fig. 9a, D was measured from both experimental specimens and simulation results under $P_0 = 3$ MPa with different k .

Fig. 9b shows that the simulated results agree well with the experimental data. As k decreases from 0.80 to 0.28, both results exhibit the same decreasing trend in D , indicating that the model can capture the effect of k on axial corner filling. The small overall deviation confirms the reliability of the finite element model, supporting its use for subsequent process parameter adjustment, instability analysis and deformation behavior investigation.

4. Results and discussion

4.1. Process window determination for complex tube forming

Using a complex tubular component of an electric-assisted bicycle frame as a representative case, the feasible process window of GLLF is determined using the proposed volume-pressure coupled finite element method. Since complex thin-walled tubes contain both circumferential and axial fillet features with different forming mechanisms, the process windows for circumferential instability suppression and axial fillet filling are investigated separately.

4.1.1. Suppression of circumferential instability

In tube GLLF, circumferential features are formed by bending followed by compression. Since bending alone is limited in forming small fillet radii, compression is required but may induce wrinkling. Therefore, the internal pressure evolution path must be properly controlled.

Based on the process window determination procedure proposed in Section 3.1, the effects of gas-phase volumetric fraction k and initial pressure P_0 on circumferential instability are analyzed to establish a process window for wrinkle suppression.

The tube is made of AA6061 ($\sigma_0 = 82.8$ MPa), with an outer diameter of 70 mm, thickness of 2.2 mm, and length of 600 mm. The P_0 is limited to ≤ 5.2 MPa to avoid yielding-induced bulging. The cavity volume decreases from 2.18 L to 1.74 L (compression ratio ≈ 0.2), and k is set above 0.24 to ensure enough compressibility.

As shown in Fig. 10a, b, under a $P_0 = 1.0$ MPa, decreasing k significantly increases the final pressure P and improves stability. Severe wrinkling occurs at $k = 0.80$ ($P \approx 1.4$ MPa), is partially suppressed at $k = 0.4$ ($P \approx 2.4$ MPa), and is eliminated at $k = 0.28$ ($P \approx 5.7$ MPa). This indicates that reducing k enhances the pressure buildup in the later stage, thereby improving resistance to compressive instability.

Fig. 10c further shows that increasing the P_0 expands the allowable range of k . When P_0 is 1.5 MPa, k must be reduced to about 0.33 to suppress wrinkling. At 3.0 MPa, k can be increased to approximately 0.72. When P_0 reaches 3.2 MPa, stable forming is achieved even at $k = 1$. These results establish the process window for wrinkle suppression, with the upper region representing instability and the lower region indicating stable forming.

Fig. 10d shows that the required final pressure decreases with increasing P_0 , from about 4.8 MPa at 2.0 MPa to 4.2 MPa at 3.2 MPa. This demonstrates that wrinkling is governed not only by the final pressure but also by the pressure path: insufficient early pressure promotes instability, while higher initial pressure suppresses it at the source.

Overall, the process window for suppressing circumferential instability is determined by the coupled influence of k and P_0 , providing a quantitative basis for stable GLLF.

4.1.2. Suppression of axial corner underfilling

In GLLF of complex thin-walled tubes, axial features are strongly influenced by the evolution of internal pressure due to pronounced geometric variations along the axial direction. This section focuses on axial transition fillets and, based on process window determination procedure proposed in Section 3.1, investigates the effects of P_0 and k on die-fitting behavior, thereby establishing the corresponding process window.

As illustrated in Fig. 11a, b, with $P_0 = 3$ MPa, the pressure response and die-fitting quality can be effectively controlled by adjusting k . When $k = 0.80$, the terminal pressure is about 4.2 MPa, and significant underfilling occurs in the axial fillet region. As k decreases to 0.35, the final pressure increases to approximately 8.9 MPa, and the die-fitting condition is notably improved. When k is further reduced to 0.28, the final pressure reaches about 18.2 MPa, and complete die fitting is achieved. This indicates that reducing k enhances the pressure buildup in the later stage and improves the bulging capacity of axial features.

The influence of P_0 is further analyzed in Fig. 11c, d. When P_0 is 1 MPa, k must be reduced to about 0.245 to achieve full die fitting. As P_0 increases to 4 MPa, k can be increased to 0.30, and when P_0 reaches

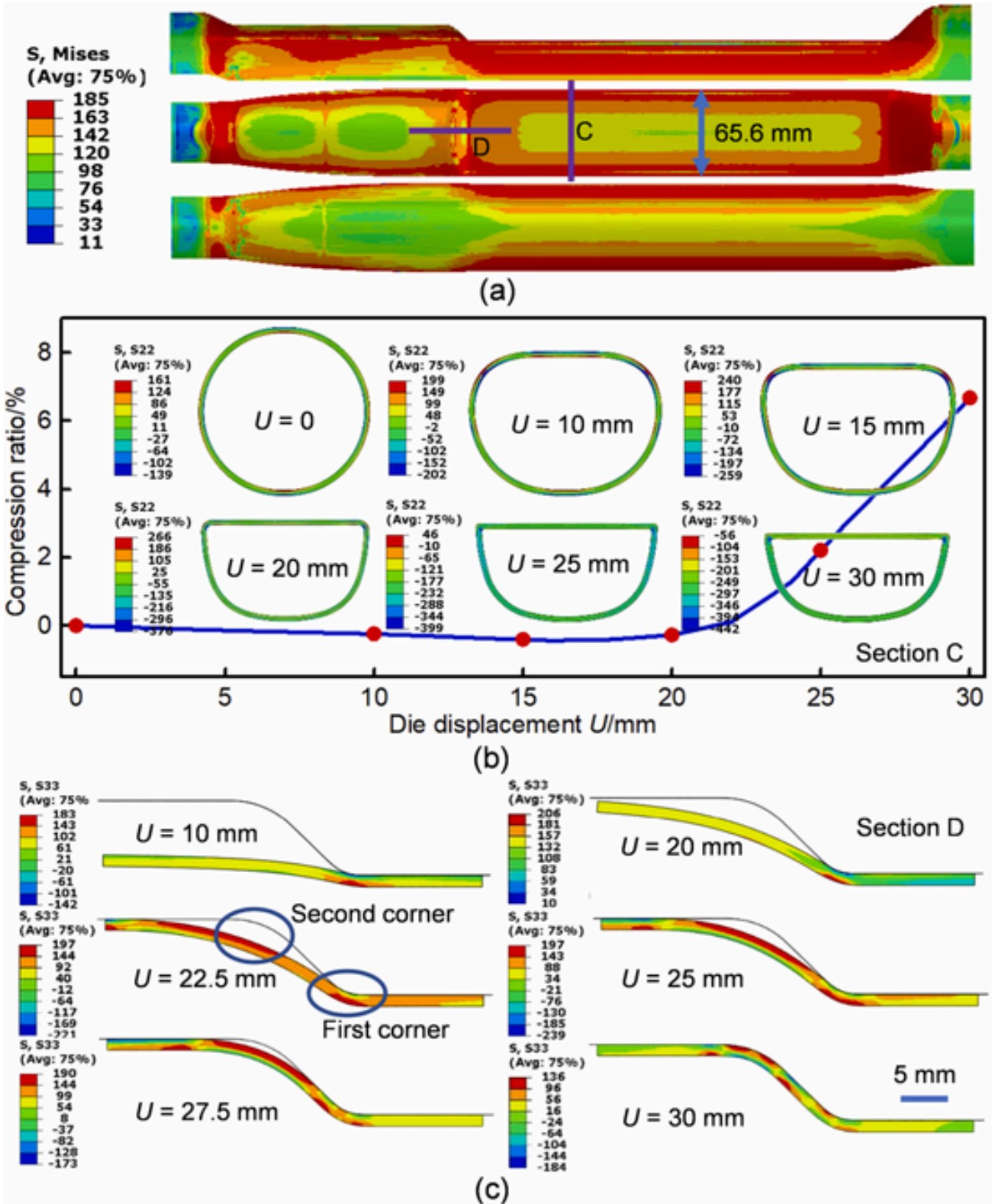


Fig. 14. Stress distribution and deformation evolution during forming: (a) Overall von Mises stress distribution at different forming stages; (b) Circumferential stress S22 (MPa), cross-sectional shape evolution and compression ratio at section A with increasing die displacement; (c) Axial stress S33 (MPa) distribution and deformation evolution in the transition region.

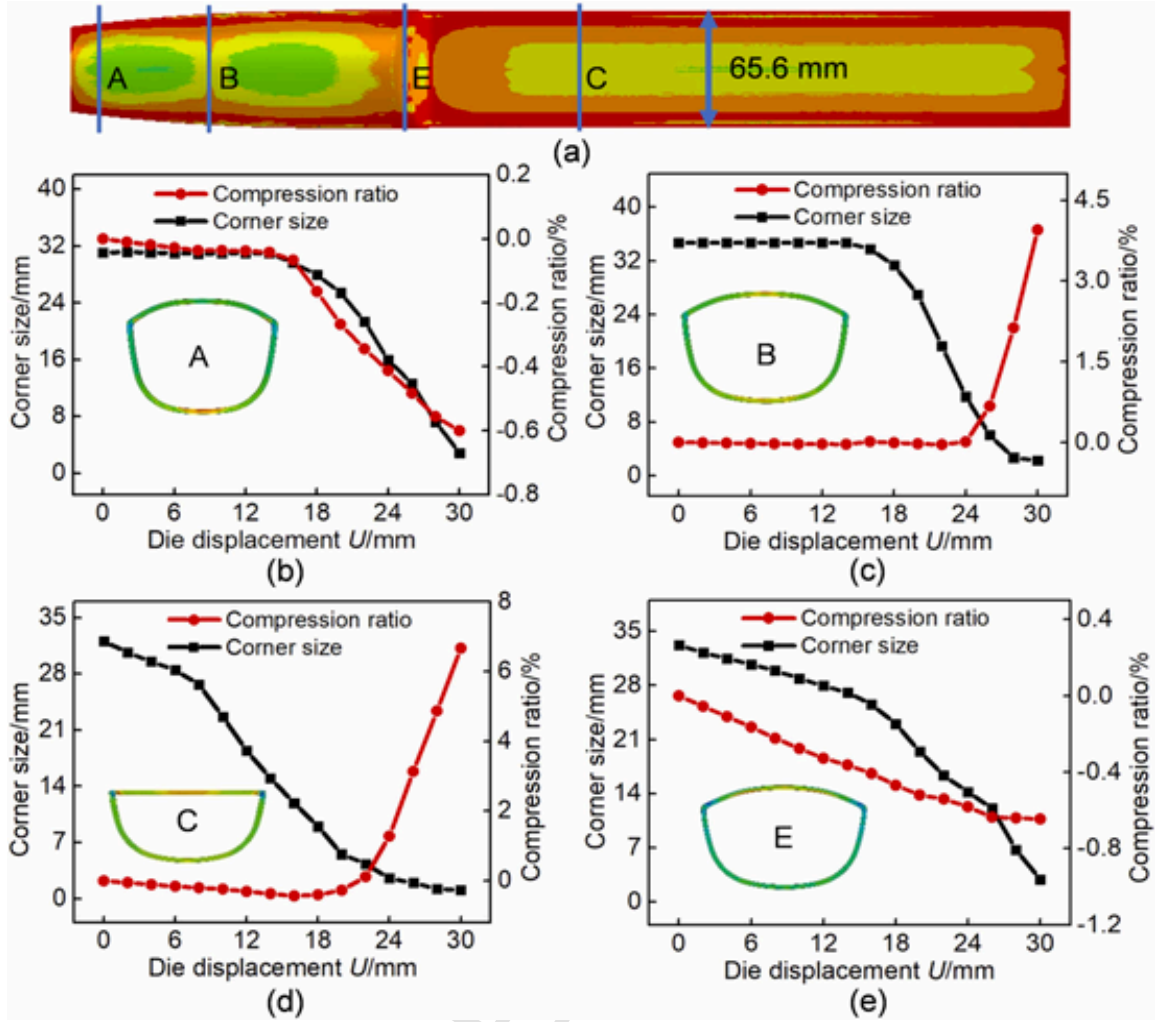


Fig. 15. Evolution of compression ratio and corner radius at typical cross-sections during forming: (a) Distribution of typical sections (A-E) along the tube; (b-e) Evolution of corner size and circumferential compression ratio with die displacement at sections A-E, respectively.

5 MPa, k can be further relaxed to approximately 0.325. These results define the critical relationship between P_0 and k , establishing the process window for complete die fitting, with the upper region representing underfilling and the lower region indicating stable forming. Increasing P_0 clearly broadens the feasible range of k and improves process robustness.

Fig. 11e shows that the die-fitting behavior is primarily governed by the peak P_0 . Once the peak pressure exceeds a critical value, complete die fitting can be achieved, and this critical pressure decreases with increasing P_0 . This suggests that higher P_0 not only improves early-stage stability but also reduces the required peak pressure by adjusting the pressure path.

In summary, axial feature forming is controlled by the internal pressure evolution. The proposed method identifies the critical k under different P_0 and establishes the process window for complete die fitting, enabling stable forming while avoiding collapse defects.

4.1.3. Experimental validation

This section experimentally validates the process determination results using AA6061 complex thin-walled tubes. Since axial feature forming requires a higher internal support pressure, the parameter set determined for axial fillet filling was selected as the reference condition. With P_0 fixed at 3 MPa, k is varied to regulate pressure evolution,

investigate its effect on axial fillet filling, and verify the reliability of the method.

As shown in Fig. 12, the axial feature in the transition zone improves progressively with decreasing k (i.e., increasing P_0). At a high $k = 0.80$, the final pressure is low, leading to severe collapse and poor die fitting in both axial and circumferential directions. As k decreases, the internal pressure increases, the collapse region is reduced, and a clearer fillet profile gradually forms. When k is reduced to 0.28, the final pressure reaches approximately 18 MPa, and complete die fitting is achieved, meeting the target geometry.

To quantitatively evaluate forming accuracy, the transition zone is scanned in 3D and the deviation is analyzed, as shown in Fig. 12c. At low pressure, the overall die fitting is poor, with a maximum deviation of about 2.6 mm at $k = 0.80$ (final pressure ≈ 4 MPa). When k decreases to 0.35 (final pressure ≈ 9 MPa), the maximum deviation is reduced to less than 1 mm. Further reducing k to 0.28 (final pressure ≈ 18 MPa) results in a maximum deviation below 0.05 mm, indicating complete die fitting. These results demonstrate that increasing internal pressure not only suppresses local collapse but also significantly improves overall forming accuracy.

Overall, axial feature forming is highly sensitive to internal supporting pressure, which directly determines die-fitting quality and dimensional accuracy. As the pressure increases, the forming state evolves from collapse to full die fitting, consistent with the numerical predic-

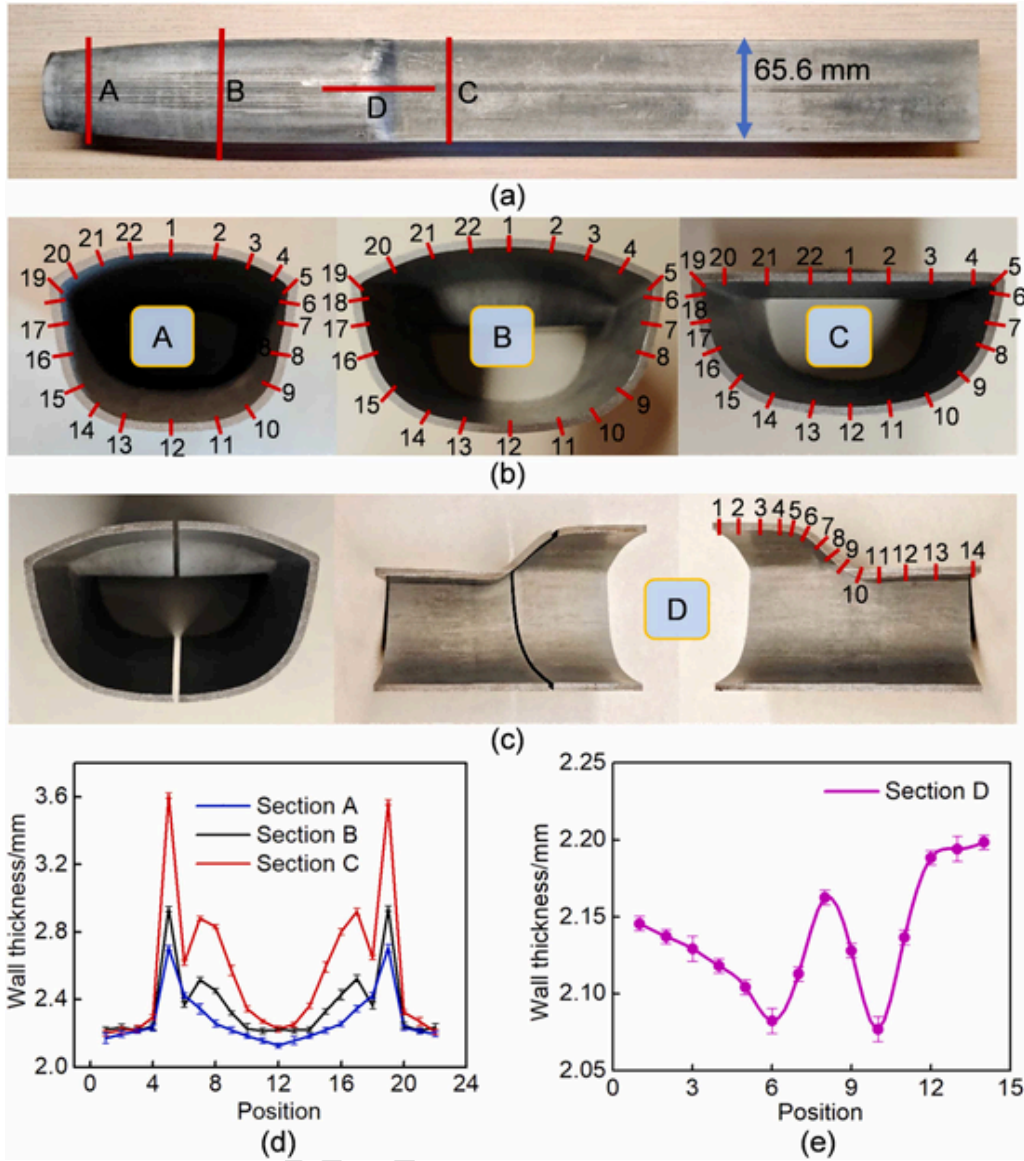


Fig. 16. Wall thickness measurement locations and distribution results of the complex thin-walled tube: (a) Formed component and selected axial sections A–D; (b) Cross-sectional geometries and measurement points of sections A–C; (c) Axial mid-section D and its measurement points; (d) Wall thickness distributions of sections A–C; (e) Wall thickness distribution along section D.

tions. This confirms that the proposed FE-based inverse determination method for GLLF process parameters has strong predictive capability and practical applicability, providing reliable guidance for high-quality forming of complex thin-walled tubes.

4.2. Stress and deformation analysis of complex thin-walled tubes

This section analyzes the stress distribution and deformation behavior in tube GLLF, aiming to clarify the deformation mechanisms and material flow characteristics of complex thin-walled tubes. First, the deformation mechanisms are identified based on the stress distribution. Subsequently, the material flow and deformation evolution are further investigated through detailed deformation analysis.

4.2.1. Stress analysis of complex thin-walled tubes

To further elucidate the mechanical mechanisms governing cross-sectional evolution and local forming during tube compression, the stress distribution was analyzed from both global and local perspectives.

As shown in Fig. 13, by assigning the material orientation of the tube blank in a cylindrical coordinate system, S_{11} , S_{22} , and S_{33} correspond to the thickness, circumferential, and axial stresses, respectively. Fig. 14 shows the overall stress distribution, the circumferential stress S_{22} distribution at section A at different forming stages, and the axial stress S_{33} distribution in the transition region, respectively.

As shown in Fig. 14a, the stress distribution during forming is highly non-uniform. Section A exhibits the highest circumferential compression ratio and overall stress level, with the maximum stress concentrated in the fillet region, indicating that it is the primary deformation zone. The high-stress region propagates from the fillet to the sidewalls and top/bottom surfaces, suggesting that deformation initiates at the fillet and gradually transfers to adjacent regions rather than occurring uniformly.

Fig. 14b shows that the circumferential stress evolution at section A exhibits clear stage-dependent characteristics. In the early stage, bending deformation dominates, with the fillet region under inner compression and outer tension, while the straight wall exhibits outer



Fig. 17. Customized GLLF system and formed tubular components: (a) Overall equipment layout, including the control module, forming module, forming die, gas-filling unit and liquid filling unit; (b) Mass-produced complex thin-walled tubular components formed using the proposed GLLF process.

compression and inner tension. As the forming proceeds and the compression ratio increases, the stress distribution gradually transitions from localized tension–compression alternation to overall compressive stress, with the maximum compressive stress near the fillet. Meanwhile, the straight wall also evolves from a bending-dominated stress state to a compressive state, and eventually the entire circumference is subjected to compressive stress. This indicates that the transformation from a circular to a square cross-section evolves from bending-dominated to compression-dominated deformation.

Fig. 14c reveals a sequential forming characteristic in the axial transition region. The first fillet forms earlier and is dominated by bending deformation, characterized by outer tension and inner compression, with tensile stress in the adjacent region indicating material flow toward the fillet. After its formation, the second fillet begins to deform, initially dominated by tensile stress on both inner and outer layers, indicating bulging-controlled deformation. As deformation progresses, the stress state transitions to outer tension and inner compression, while the surrounding region exhibits outer compression and inner tension, suggesting that the formation of the second fillet results from the combined effects of bending and stretching.

Overall, the fillet region remains the dominant zone of stress concentration and deformation throughout the forming process. The circumferential deformation at section A evolves from bending-dominated to compression-dominated behavior, while the axial transition region exhibits sequential forming characteristics. The formation of complex cross-sections is essentially governed by the coupled effects of local bending, overall compression, and axial material flow.

4.2.2. Deformation analysis of complex thin-walled tubes

This section further investigates the deformation behavior in tube GLLF. Since the process is dominated by circumferential bending and compression, four representative cross-sections along the axial direction are selected to analyze the evolution of compression ratio and fillet radius size during shape transformation.

As shown in Fig. 15, both the compression ratio and fillet radius exhibit clear stage-dependent evolution with increasing die displacement. For section C (Fig. 15d), the maximum circumferential compression ratio reaches approximately 7%. In the early and intermediate stages, the compression ratio remains nearly constant and the perimeter is unchanged, indicating bending-dominated deformation. In the later stage, the compression ratio increases rapidly and the perimeter begins to shrink, indicating a transition to compression-dominated deformation. The fillet radius decreases gradually in the early stage, rapidly in the intermediate stage, and stabilizes in the later stage. When the fillet radius reduces to about 4.3 mm, compression initiates, indicating a bending limit. Further compression leads to a stable radius of approximately 1.2 mm, suggesting a compression limit.

For section B (Fig. 15c), the maximum compression ratio is about 4%. No compression occurs when the die displacement is below 24 mm,

while compression increases linearly beyond 26 mm. The fillet radius remains unchanged before contact and decreases rapidly to about 6 mm after contact, indicating bending-dominated deformation. In the subsequent stage, it decreases gradually to about 2.3 mm under compression. Sections A and E (Fig. 15b, e) exhibit negligible compression. After die contact, the fillet radius decreases gradually, while the perimeter slightly increases (up to about 0.6%), indicating tensile effects during bending.

Overall, the forming process can be divided into three stages: free (non-contact), bending, and compression. The bending stage is experienced by all sections and is characterized by constant perimeter and rapid reduction of fillet radius. The compression stage occurs only in certain sections, featuring perimeter shrinkage and further reduction of fillet radius. The presence of the free and compression stages depends on the initial geometry. As a result, deformation among different axial sections is inherently non-synchronous, posing additional challenges for process parameter design.

4.3. Wall thickness characteristics of complex thin-walled tubes

This section investigates the wall-thickness characteristics of AA6061 complex thin-walled tubes. Qualified formed components are sectioned for thickness measurement at three representative circumferential sections A–C with different geometries and compression ratios, as well as at the axial mid-section D. As shown in Fig. 16, the circumferential measurement points are symmetrically arranged on the upper surface, corner regions, sidewalls and bottom surface.

The effects of cross-sectional shape and compression ratio on wall thickness distribution are first analyzed. As shown in Fig. 16d, section A exhibits negligible circumferential compression, and the wall thickness at the top and bottom surfaces remains nearly unchanged. However, due to bending deformation, slight thickness variations occur: the corner region thickens due to reduced curvature radius, while the flat regions show slight thinning, indicating a bending-dominated behavior. With increasing compression at section B, circumferential compression becomes significant, leading to overall thickening, particularly in the fillet region (maximum 2.9 mm) and sidewall (2.5 mm), while the top and bottom surfaces remain relatively unchanged. A local thickness decrease near the fillet indicates non-uniform material flow and strain redistribution under coupled bending and compression. At higher compression (section C, ~7%), overall thickening is further enhanced, with the maximum thickness reaching 3.6 mm. Meanwhile, slight thinning still occurs near the fillet, suggesting that material accumulation toward high-curvature regions is accompanied by localized stretching. Overall, the wall thickness increases gradually from the lower region toward the fillet, reflecting continuous material flow and accumulation along the circumferential direction.

Fig. 16e shows the wall thickness distribution in the axial S-shaped transition region. The results reveal a clear sequential forming

behavior. The lower fillet (position 10) forms first under die loading, dominated by bending and stretching, resulting in localized thinning with limited influence. Subsequently, the upper fillet (position 6) forms under internal pressure, dominated by coupled bulging and bending, leading to more pronounced thinning in the fillet center and surrounding areas. The differences between the two fillets indicate that loading mode and forming sequence significantly affect the strain path, thereby governing the wall thickness distribution.

Overall, the wall thickness evolution of complex thin-walled tubes is governed by the coupled effects of bending, circumferential compression, and bulging. The compression ratio controls the overall thickening level, geometric curvature determines local thickness variation, and the forming sequence influences material flow paths and thickness non-uniformity. Therefore, process design should coordinate compression and loading paths, ensuring enough material supply to fillet regions to achieve uniform thickness distribution and improved structural performance while avoiding excessive thickening or thinning.

5. Application

As shown in Fig. 17, a customized GLLF system was developed to implement the proposed volume–pressure coupled process design method for complex thin-walled tubular components used in electric-assisted bicycle frames. The system integrates liquid supply, gas supply, forming and control modules, enabling coordinated regulation of liquid volume, initial gas pressure and die closing displacement.

The application value of the proposed method lies in mechanism-based parameter determination. By adjusting the gas–liquid ratio and initial pressure, the compressibility of the internal medium and the resulting volume-induced pressure response can be regulated during die closing. This adaptive pressure support helps suppress circumferential compression instability, improve axial corner filling and adjust wall thickness distribution. Based on the coupling among cavity volume evolution, gas–liquid mixture compression and pressure response, the effects of key parameters on material flow and forming quality can be quantitatively evaluated before experimental trials, thereby reducing process uncertainty and improving forming consistency.

Since this coupling mechanism is not limited to a specific geometry, the method provides transferable support for process design and deformation control of complex thin-walled tubular components with different cross-sections, local fillets and axial features. It has therefore been applied to the forming of dozens of tubular components, shortening product development cycles and improving processing efficiency.

6. Conclusions

In tube GLLF, cavity volume variation and gas–liquid mixture compression induce nonlinear pressure responses that affect instability suppression and deformation behavior. However, effective methods for analyzing this volume–pressure coupling remain insufficient, and the axial and circumferential deformation mechanisms in tube GLLF are still unclear, limiting process-parameter design and forming-quality control. To address these issues, a FE-based process design and determination method incorporating dynamic volume–pressure coupling was proposed and experimentally validated. The principal conclusions are summarized as follows:

(1) A finite element method was developed for GLLF process design and process window determination of complex thin-walled tubes by integrating restart analysis with a gas–liquid mixture compression model. The key contribution is the bidirectional coupling between cavity-volume variation, pressure response, and deformation evolution, where deformation-induced volume change governs pressure evolution, and the updated pressure regulates subsequent deformation. Based on this mechanism, the initial pressure and gas-phase volumetric fraction were

determined to suppress circumferential instability and improve axial corner filling, thereby defining the process window for stable forming.

(2) For the studied component, axial corner filling, governed by higher pressure demand, was identified as the critical design constraint and used to experimentally validate the proposed method. The results demonstrate that pressure evolution controls the transition from insufficient support and incomplete filling to stable deformation and complete die fitting. At $P_0 = 3$ MPa, reducing k from 0.80 to 0.28 increased terminal pressure from about 4 MPa to 18 MPa and reduced maximum corner deviation from 2.6 mm to below 0.05 mm, showing good agreement between simulation and experiment.

(3) Stress and cross-sectional deformation analyses clarified the deformation mechanism of typical sections and quantified the evolution of corner-radius formation. The sections experienced initial deformation, bending, and compression. Bending dominated the main shape transformation under an approximately constant perimeter, converting the initial circular tube with a radius of 35 mm into complex sections with transitional corner radii of 4–6 mm. Subsequent compression overcame the bending-forming limit, reducing the radii to 2.3 mm at 4% compression and 1.2 mm at 7%, confirming the effectiveness of compression-assisted forming for small fillet radii.

(4) Wall-thickness analysis revealed the redistribution mechanism governed by curvature change, compression ratio, and deformation sequence. In circumferential sections, wall thickening occurred in regions of decreasing curvature. The overall thickness level was mainly controlled by compression ratio, while the local distribution depended on geometric complexity. In axial sections, earlier-deforming regions mainly experienced bending-induced thinning, whereas later-deforming regions were more susceptible to thinning caused by deformation accumulation and unsupported bulging. For the S-shaped section, such bulging-induced thinning led to a slight thickness reduction from 2.2 mm to 2.15 mm.

Although the proposed method provides an effective approach for process window determination and deformation control in tube GLLF, several issues remain to be addressed. Future work should focus on developing continuous volume–pressure response and real-time pressure-control strategies for components with small cavity-volume variation, independently investigating instability mechanisms under compression-dominated deformation, and extending this high-efficiency forming method to more complex tubular structures and broader industrial applications.

CRedit authorship contribution statement

Zhubin He: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Hang Yuan:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Junzhuo Wang:** Writing – review & editing, Investigation, Formal analysis. **Yanli Lin:** Writing – original draft. **Zhijie Hu:** Validation, Data curation. **Haochen Zhao:** Software. **Gaoning Tian:** Visualization, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data availability

Data will be made available upon request.

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